MACHINE LEARNING WEEK 14

LAB REPORT

Nevin Mathew Thomas

PES2UG23CS381

Course: UE23CS352A

20-11-2025

1. Introduction:

The objective of this lab was to design, build, train, and evaluate a Convolutional Neural Network (CNN) using PyTorch to classify images of hand gestures into three categories: rock, paper, and scissors. The dataset was automatically downloaded from Kaggle and organized using PyTorch's ImageFolder utility to assign labels based on directory structure. All images were preprocessed by resizing them to 128×128 pixels and applying normalization to ensure consistent input to the network. The completed model was trained on augmented batches of data to learn meaningful visual features and then evaluated on a held-out test set to measure its classification accuracy and overall performance.

2. Model Architecture:

The CNN model consists of three convolutional blocks, each followed by a ReLU activation and MaxPooling:

Convolutional Layers:

1. Conv Block 1
    - Conv2d: 3 → 16 channels
    - Kernel size: 3×3
    - Padding: 1
    - Activation: ReLU
    - Max Pooling: 2×2 (reduces 128 → 64)
2. Conv Block 2
    - Conv2d: 16 → 32 channels
    - Kernel size: 3×3
    - Padding: 1
    - Activation: ReLU
    - Max Pooling: 2×2 (64 → 32)
3. Conv Block 3
    - Conv2d: 32 → 64 channels
    - Kernel size: 3×3
    - Padding: 1
    - Activation: ReLU
    - Max Pooling: 2×2 (32 → 16)

After the final pooling layer, the image size becomes 16 × 16, and the feature map dimension is 64 × 16 × 16.

Fully Connected Classifier:

The classifier contains:

- Flatten Layer
- Linear Layer: 64×16×16 → 128
- ReLU Activation
- Dropout: 0.3 (prevents overfitting)
- Final Linear Layer: 128 → 3 (one neuron per class)

This architecture provides a balance between simplicity and enough capacity to learn hand gesture features such as edges, contours, and shapes.

3. Training and Performance:

Hyperparameters Used:

- Optimizer: Adam
- Loss Function: CrossEntropyLoss
- Learning Rate: 0.001
- Batch Size: 32
- Train/Test Split: 80% / 20%
- Epochs: 10
- Image Size: 128×128
- Normalization: mean = (0.5, 0.5, 0.5), std = (0.5, 0.5, 0.5)

Test Accuracy:

```
Test Accuracy: 99.09%
```

4. Conclusion and Analysis:

The CNN performed well on the Rock–Paper–Scissors dataset, achieving a high test accuracy of 99.09%. This shows that the model effectively learned relevant visual features from the images. The training process was stable, and both loss reduction and accuracy progression behaved as expected.

Possible Improvements:

1. **Data Augmentation**
   Add transforms such as random rotations, flips, and color jitter to improve robustness and reduce overfitting.
2. **Deeper or More Advanced Models**
   Use architectures like ResNet-18, MobileNet, or VGG-like CNNs to further increase accuracy.
3. **Learning Rate Scheduling**
   Use schedulers such as StepLR or ReduceLROnPlateau for more refined optimization.
4. **Train for More Epochs**
   Increasing from 10 to 20–30 epochs could boost accuracy further.